

DENVar: A Global Dynamic Estimate of NDVI Variance

Stefano Mezzini^{1,2,3}, Gavin L. Simpson⁴, and Michael J Noonan^{1,2,5,*}

¹Okanagan Institute for Biodiversity, Resilience, and Ecosystem Services, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

²Department of Biology, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

³Poisson Consulting Ltd., Nelson, British Columbia, Canada.

⁴Department of Animal & Veterinary Sciences, Aarhus University, Denmark.

⁵Department of Computer Science, Math, Physics, and Statistics, The University of British Columbia Okanagan, Kelowna, British Columbia, Canada.

*Address correspondence to: michael.noonan@ubc.ca

Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and analyses. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance (i.e., stochasticity) in NDVI has gained more attention, particularly as climate change and anthropogenic activities transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean, which depends on the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*, along with corresponding estimates of mean NDVI. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. Globally, both mean NDVI and *DENVar* varied substantially spatiotemporally, but both indices had strong pulses globally in 1990 and 1996. Following the 2015-2016 El Niño event, both indices showed strong latitudinal stratification saw a widespread increase except near the Equator and high northern latitudes. Across biomes, *DENVar* often had a concave-down relationship with mean NDVI, and it had complex nonlinear relationships with other common environmental variables, including temperature, precipitation, and species richness. We conclude by discussing some of the potential applications of *DENVar* and how it relates to some fundamental theories in ecology, such as optimal foraging theory, regime shifts, and Taylor's power law. We provide a publicly and freely available repository for downloading rasters of daily estimates and long-term-averages of *DENVar* and the related estimates of mean NDVI.

1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology (Pettorelli et al., 2005, 2011), landscape phenology (Merkle et al., 2016; Hmimina et al., 2013), agriculture (Franch et al., 2017), forestry (João et al., 2018), and conservation (Mallegowda et al., 2015; Marcus et al., 2025). High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork (Bhatti et al., 2024; Borowik et al., 2013) and leverage such estimates for studying large-scale plant dynamics (Pettorelli et al., 2005) and their effects on animal movement behavior (Merkle et al., 2016; Aikens et al., 2017). Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades (Levins, 1974; Stephens and Charnov, 1982; Lande, 1993; Sæther, 1997), but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms (Mezzini et al., 2025; Rizzuto et al., 2021) to landscapes (Zhang et al., 2009; Mutti et al., 2020; Marcus et al., 2025). However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals’ perception of and ability to respond to change (Levin, 1992). A forest fire is severely disruptive and deadly for many insects (Fettig et al., 2022), while birds and large mammal may not struggle to escape the fire and or establish a new range (Franklin et al., 2021; Calhoun et al., 2024). Similarly, the effects of the fire recovery will depend on its duration relative to other organisms’ phenology and lifespan (Frankham and Brook, 2004). Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce species’ ability to adapt or specialize (Chevin et al., 2010), alter species’ movement behavior (Mezzini et al., 2025), and even affect extinction risk (Lande, 1993; Bull et al., 2007; Fung et al., 2018). Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied (Marcus and Noonan, 2023). Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI while also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement of primary productivity but a proxy, it should be interpreted carefully (Pettorelli et al., 2011). NDVI is prone to sensitivity loss in very green environments (Matsushita et al., 2007; Son et al., 2014), values vary across sensors (Beck et al., 2011; Fan and Liu, 2016; Huang et al., 2021), and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends (Goodchild and Quattrochi, 2023; Reid et al., 2018). However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particularly convenient due to their ability to learn flexibly from data (Wood, 2017) and handle very large spatiotemporal datasets (Wood et al., 2015, 2017).

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate

of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors, and GAMs. Although satellites with AVHRR and VIIRS sensors provide data at a coarser spatial resolution (0.05°) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opacity of machine learning models such as neural networks (Simpson, 2018; Wood, 2017). We illustrate the relationships between DENVar and multiple relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

2 Methods

2.1 Datasets used

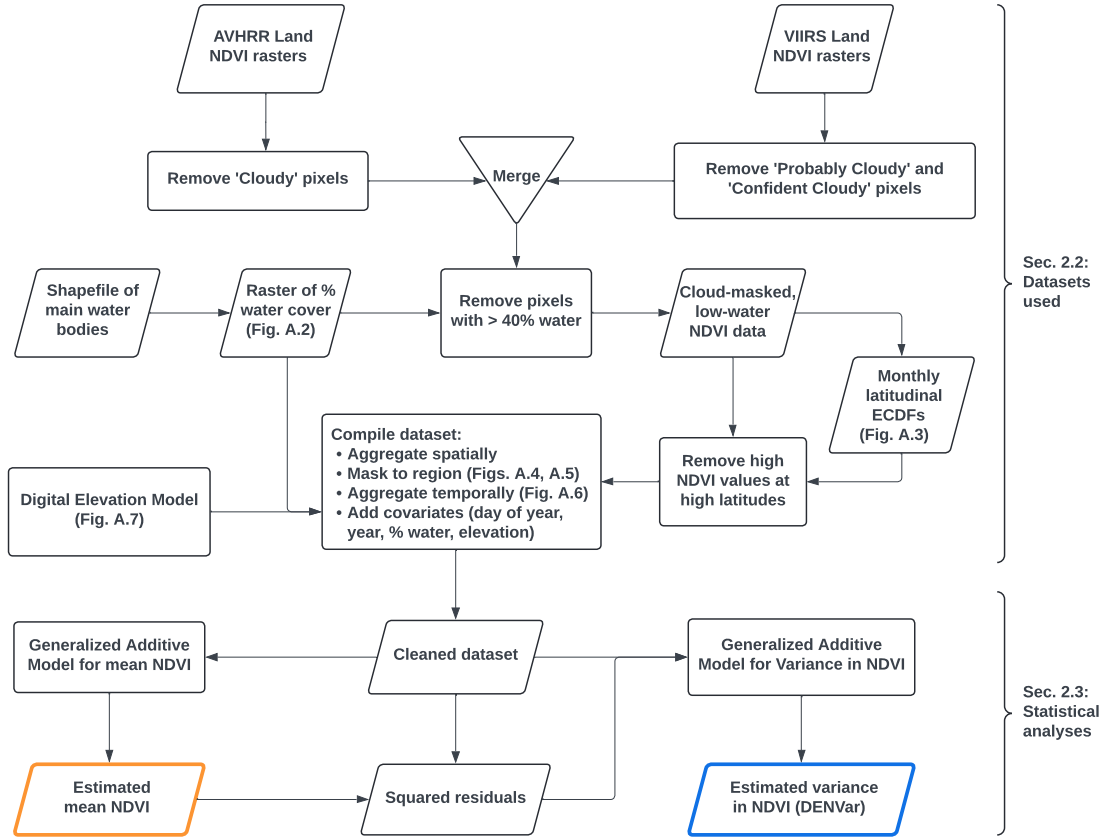


Figure 1: **Flowchart for the analyses used in the manuscript.** The curly braces on the right indicate the subsections in the Methods section that detail the dataset collation and cleaning (2.1), and statistical analyses (2.2).

Fig. 1 illustrates the processes of data collection, cleaning, and modeling used in this project. We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; United States National Centers for Environmental Information (NCEI) and United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2018) and VIIRS sensors (2014-01-01 to 2025-06-29; United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2022).

We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS rasters. To remove pixels that included large portions of water, we used a shapefile of major water bodies (Garmin International, Inc. and Esri, 2025) to produce a static raster of the percentage of water within each pixel, at the same spatial resolution as the NDVI rasters (0.05° ; Fig. A.2). We removed all pixels with $> 40\%$ water cover.

Many rasters had horizontal bands of $\text{NDVI} > 0.9$ at unrealistically northern latitudes between September and March, likely due to limited visibility during fall and winter in the Northern hemisphere. We identified and removed problematic pixels using the monthly empirical cumulative distribution functions of NDVI across latitude (Fig. A.3). We did not filter problematic values at latitudes $< 60^\circ\text{S}$ since we fully excluded Antarctica from the analyses (Fig. A.4).

To ensure the number of rows for each dataset was below the maximum data frame size in R ($2^{31} - 1 \approx 2.1 \times 10^9$), we reduced the dataset size by splitting it across four spatial groups (Figs. A.4 and A.5), and aggregating the rasters spatially and temporally by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of $0.05^\circ \times 4 = 0.2^\circ$ (Fig. A.6). Finally, we annotated the dataset with columns of day of year (1-366), integer year (1981-2025), percent water cover, and elevation above sea level (m). The digital elevation model was obtained using the `get_elev_raster()` function from the `elevatr` package for R (v. 0.99.0; Hollister et al., 2023) with an original resolution of 0.076° (Fig. A.6). Despite the substantial reduction in dataset size, the dataset creation was still challenging for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

2.2 Statistical Analyses

We estimated spatiotemporal trends in mean and variance in NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3 Wood, 2017). To reduce modeling fitting times with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE`; Wood et al., 2015, 2017). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation $Y^* = (Y + 1)/2$; Merder et al., 2026) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests and work by Marcus et al. (2025) showed that beta GAMs and particularly beta location-scale GAMs had prohibitive requirements for both fitting times and memory requirements, since both requirements scale $O(Mnp^2)$, for M smoothing parameters and a model matrix with n rows and p columns (Wood et al., 2015). A

124 beta location-scale model for only the island of Sardinia (24,000 km²; Appendix B) was interrupted
 125 after 144 hours, during which the model had only completed three steps using the Newton method
 126 described by Wood and Fasiolo (2017). In contrast, Gaussian models with fixed scaled parameters
 127 (i.e., constant variance) fit substantially faster and provided very similar spatiotemporal estimates
 128 of mean and variance in NDVI (Appendix B). Thus, we estimated trends in mean and variance in
 129 NDVI using a pair of GAMs for each region shown in Fig. A.4; one for estimating mean NDVI over
 130 space and time (assuming a constant variance), and one for estimating the mean squared residual
 131 (i.e., variance) in NDVI (again, assuming a constant scale parameter in each model).

132 Together, the sets of models for mean NDVI (i.e., $\mu(u, t)$) and variance in NDVI (i.e., $\sigma^2(u, t)$)
 133 for location u and time t had the following structure, with $\nu(u, t)$ indicating NDVI:

$$\left\{ \begin{array}{l} \nu(u, t) \sim \text{Normal}(\mu(u, t), \sigma^2(u, t)) \\ \mu(u, t) = \mathbb{E}(\nu(u, t)) \approx \hat{\mathbb{E}}(\nu(u, t)) = \hat{\mu}(u, t) \\ \hat{\mu}(u, t) = \beta_{\mu} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) \\ \sigma^2(u, t) = \mathbb{E}((\nu(u, t) - \mu(u, t))^2) \approx \hat{\mathbb{E}}((\nu(u, t) - \hat{\mu}(u, t))^2) = \hat{\sigma}^2(u, t) \\ \hat{\sigma}^2(u, t) = \beta_{\sigma^2} + s(\text{prop_water}) + s(\text{elev_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) + s(\hat{\mu}(u, t)) \end{array} \right., \quad (1)$$

134 where β_{μ} and $\beta_{\sigma^2(u, t)}$ are model intercepts, and, for a given (spatiotemporally aggregated) pixel:
 135 **prop_water** is the proportion of water within the pixel, **elev_m** is elevation above sea level in meters,
 136 **doy** is integer day of year (1-366), **year** is integer year (1981-2025), and **lat** and **long** are latitude
 137 and longitude, respectively. We omit the pixel-level indices for ease of readability. $f(\cdot)$ indicates
 138 smooth functions, which were estimated using **mgcv**'s **s()** function for main effects (i.e., only one
 139 variable in the parentheses) or a decomposition into main effects (**s()**) and interaction terms via
 140 pure tensor product interactions **ti()** to avoid including the main effects more than once (Wood,
 141 2017). Seasonal terms used cyclical cubic splines (**bs** = 'cc') with edge knots 0.5 and 366.5, while
 142 spatial smooths used a basis with second-order spline on the sphere (**bs** = 'sos'; Wood, 2017). The
 143 linear predictors for $\mu(u, t)$ and $\sigma^2(u, t)$ have the same structure, other than the $f(\mu(u, t))$ smooth
 144 function for $\sigma^2(u, t)$, but the functions for $\sigma^2(u, t)$ are independent of those for $\mu(u, t)$.

145 An interaction term of **doy**, **year**, and space was not included in the model due to the already
 146 high computational costs and long fitting times. Despite the data aggregation procedures and
 147 faster fitting methods, including parallelizing computation across 60 cores, each of the four models
 148 required multiple days to fit. Predicting from the models for $\hat{\mu}(u, t)$ also had substantial time and
 149 memory demands, since predictions were calculated without discrete prediction methods (**discrete**
 150 = **FALSE**). While discrete predictions were orders of magnitude faster, they were excessively spatially
 151 coarse, with a spatial resolution of approximately $1^\circ \times 1^\circ$. The spatial resolution depended on a
 152 combination of the area modeled and the basis size for the spatial terms. Fine-scale predictions

(i.e. `discrete = FALSE`) for the Neotropic-Nearctic region required 40 days despite consistently using > 1 TB of RAM. See the code availability statement for additional information. See Wood et al. (2017) and Wood et al. (2015) for additional information on the discretized approximation of the smooth terms.

From the two sets of models, we produced daily estimates of (1) the Dynamic Estimate of NDVI Variance (DENVar) and (2) the estimated mean NDVI they are conditional on, at a 0.2° resolution. The rasters are freely available on GitHub (see code availability statement), along with yearly and long-term (1981-2025) averages of both DENVar and the estimated mean NDVI. Since there are no corrections to any bias of $\hat{\mathbb{E}}((\nu - \hat{\mu}(u, t))^2)$, the estimates of $\sigma^2(u, t)$ may be inflated near areas or periods of rapid and substantial change. Still, the appreciability of this inflation depends on the spatiotemporal scale over which changes are perceived (Levin, 1992; Frankham and Brook, 2004; Goodchild and Quattrochi, 2023; Mezzini et al., 2025). Since estimates of variance are conditional on the corresponding estimates of the mean, we suggest users always use the corresponding rasters of mean NDVI in their analyses rather than using independent estimates of mean NDVI.

We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu* palette (Crameri, 2018a; Crameri et al., 2020), which has high-contrast for deuteranope and protanope vision (Fig. A1). We obtained all color palettes with three or more colors from the *khroma* package (v. 1.14.0 Frerebeau, 2024) for R (v. 4.4.0; R Core Team, 2025).

3 Results

The estimates of mean NDVI ($\hat{\mu}(u, t)$) and DENVar ($\hat{\sigma}^2(u, t)$) were spatially smooth but still captured complex spatial trends (Fig. 2). The models for mean NDVI explained 68-88% of the deviance, while the models for DENVar explained 13.40-18.60% of the deviance (Table 1). For both models, predictions were unreliable in areas with very scarce data (e.g., rock and ice biomes in the Arctic; see Fig. A.5), so such areas were excluded when producing figures and estimates of mean NDVI and DENVar. Still, the models detected complex spatial patterns in both variables, such as along the Andes and the Himalayas, and in the Palearctic and Nearctic taiga.

Region	Dev. expl. (Mean NDVI, %)	Dev. expl. (DENVar, %)
Africa	88.02	18.60
Indo-Malay, Oceania, Australasia	67.89	18.17
Neotropic, Nearctic	80.65	13.40
Palearctic	82.23	14.89

Table 1: Percent deviance explained by the models for mean NDVI and DENVar for each of the four regions shown in Fig. A.4.

The models also detected complex spatiotemporal trends in both mean NDVI and DENVar, with major shifts occurring proximal to the major El Niño events in 1982, 1997, and 2015 (Fig. 3; Jia et al., 2025). In terms of temporal trends, the early 1980s saw lower mean NDVI and greater variance in NDVI, with values increasing during the second half of the decade (Fig. 3C, D). In the 1990s, there were two global-scale, multiannual oscillations in both mean NDVI and DENVar

that ended with a widespread increase in mean NDVI and NDVI stochasticity in 2000 and 2001. The 2000s and 2010s were characterized by more heterogeneous and complex spatial patterns that intensified just before the 2015 El Niño event. Equatorial regions saw a decrease in mean NDVI and DENvar, while near-tropical regions generally experienced an increase in both metrics. Latitudes approximately above 60° experienced the strongest changes in both mean NDVI and DENVar, with strong spatiotemporal trend inversions near 59, 65, and 69°N.

Spatial patterns in long-term mean NDVI and DENVar varied across biomes (Fig. 4), with many biomes showing a wide range of both variables. However, DENVar and mean NDVI often had a concave down relationship with a peak in DENVar near intermediate values of NDVI (0.2-0.6).

Fig. 5 shows the relationship between DENVar and some common spatial variables. Generally, DENVar was highest with mean NDVI near 0.25, decreased with mean annual temperature, increased with seasonal temperature range, and increased with total annual precipitation. Additionally, DENVar was generally higher in environments with low: IUCN Red List species richness, proportion burned, human footprint, and risk of arboviral disease.

4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. We have shown that DENVar has a strong, nonlinear relationship with mean NDVI, climatic variables, species diversity, the impacts of fire and human activity, and risk of arboviral disease. Yet, environmental stochasticity is rarely accounted for in conservation research (Marcus and Noonan, 2023), although recent work has started drawing more attention to how it shapes ecosystems and evolution (Anderson et al., 2017; Logares and Nuñez, 2012; Mezzini et al., 2025; Oestreich et al., 2026). We have presented the relationships between DENVar and some common ecological metrics (Fig. 5), but have also highlighted some of the difficulties that come with estimating NDVI variance over time and space.

Ultimately, both NDVI and DENVar are only proxies, so their effects should be interpreted on a case-specific basis, and the choice of models is often as important as the choice of which data to include. In particular, we suggest careful use of (Hierarchical) Generalized Additive Models (Wood, 2017; Simpson, 2018; Pedersen et al., 2019) due to their ability to account for nonlinear relationships. Similar values of mean NDVI may indicate different levels of nutrient availability, growth stage, and forage density within different landscapes and across different sensors (Huang et al., 2021), and NDVI does not always correlate linearly with forage availability or quality (Johnson et al., 2018). Consequently, while DENVar often peaks at intermediate values of mean NDVI in the hexplots in Fig. 4, it remains unclear whether environmental stochasticity is highest at maximal or intermediate productivity. Taylor’s law (Taylor, 1961) suggests that environmental stochasticity should follow a positive power relationship with environmental productivity. While Taylor’s law applies to many systems (Kendal, 2004), there are multiple reasons why it may not apply directly to NDVI and DENVar.

Firstly, ephemeral and transient environments tend to have greater environmental stochasticity than stable and established ones. This is most visible in environments that frequently switch

between alternative stable states, where the frequent switching between states can increase both spatiotemporal heterogeneity and stochasticity. In addition, transitions from one state to another are often preceded by an increase in variance (Carpenter and Brock, 2006) or at least experience a spike in environmental variance during the transition (Beisner et al., 2003). Since environmental stochasticity reduces ecosystems’ energetic balances and carrying capacities (Chevin et al., 2010), populations can remain in a state of rapid and stochastic growth as they recover from recent mortality (Lande, 1993; Beisner et al., 2003; Arumugam et al., 2021). In contrast, stable environments are more likely to have smaller population oscillations and thus remain near carrying capacity in stable states, whether the environments be resource-poor or resource-rich (Beisner et al., 2003; Arumugam et al., 2021).

Secondly, NDVI and DENVar are not direct measurements but rather proxies of primary productivity and stochasticity. Therefore, since NDVI signatures vary across plant species, particularly between woody and herbaceous species (Huang et al., 2021), environmental productivity may not scale linearly with NDVI, and variance in NDVI may not need to scale monotonically with mean NDVI, either.

Thirdly, statistical theory suggests that, since terrestrial NDVI is bounded on both the lower end near -0.1 (snow and ice) and the upper end at 1 (maximally saturated green vegetation), DENVar likely to be highest when mean NDVI is far from either boundary, which allows it to vary more in either direction. This hypothesis is supported by common characteristics of biomes that experience irregular snowfall. Firstly, ecosystems with stochastic and rapid snowfall and snowmelt (e.g., boreal forests and alpine habitats) are likely to have mean NDVI that is higher than biomes that are most often covered in snow and ice (e.g., the Arctic) or rarely experience snow (e.g., the Amazon forest). Concurrently, the higher frequency of snowfall and snowmelt results in spikes in NDVI variance when NDVI may switch between near zero (snow) and well above zero (dense vegetation) within the span of days or weeks, and greater long-term variance due to interannual variation in snow phenology.

The relationship between seasonal variation and environmental stochasticity highlights the importance of distinguishing seasonality from stochasticity. Seasonality is best interpreted as a (cyclical) change in mean conditions that many organisms can learn to predict, as in the case of green-wave surfing (Middleton et al., 2018; Geremia et al., 2019). In contrast, environmental stochasticity refers to unpredictable changes, such as fires (Fettig et al., 2022; Calhoun et al., 2024), sudden snow storms (Aikens et al., 2025), and unusual seasonal phenology (Torstenson and Shaw, 2025). Riotte-Lambert and Matthiopoulos (2020) and Mezzini et al. (2025) provide theoretical frameworks for how and why resource stochasticity increases organisms’ range sizes (also see Stephens and Charnov, 1982; Rizzuto et al., 2021), while Oestreich et al. (2026) detail how environmental stochasticity affects collective information and behaviour. Highly unpredictable environments tend to have lower energetic balances and carrying capacities (Chevin et al., 2010) and higher extinction risk (Lande, 1993; Fung et al., 2018), since communities cannot rely on stable conditions. Additionally, stochastic environments reduce species’ ability to specialize, instead favoring generalist with flexible adaptations (Levins, 1974). Spatial memory is particularly important for navigating highly stochastic environments (Steixner-Kumar and Gläser, 2020; Oestreich et al., 2026), including locating dependable waterholes during droughts (Polansky et al., 2015), avoiding ecological traps (Abrahms et al., 2019), migrating (Bracis

and Mueller, 2017; Rickbeil et al., 2019), and navigating large home ranges (Falc3n-Cort3s et al., 2021; Ranc et al., 2022). As environmental productivity and stochasticity change, species will be forced to adapt and evolve, provided that changes are gradual enough (Levins, 1974).

Environmental stochasticity should be accounted for when designating new protected areas or evaluating current ones (Marcus et al., 2025). Failing to do so can limit a protected area’s effectiveness by forcing (at-risk) species into suboptimal habitats, especially if animals cannot escape adverse conditions or expand their range as needed (Ortega et al., 2023; Aikens et al., 2020). Concerningly, recent work by (Marcus et al., 2025) showed that protected areas in Canada have similar levels of environmental stochasticity to unprotected areas, highlighting the importance of incorporating a measure of environmental stochasticity in the decision process. Furthermore, the findings of Marcus et al. (2025) mirror Levins’s (1974) hypothesis that more stochastic environments tend to have lower species diversity while also having more frequent extreme temperatures. These findings, which are also supported by our results (Fig. 5), highlight that environmental stochasticity is closely linked with many habitat characteristics and can thus reshape entire communities (Logares and Nu3ez, 2012; Oestreich et al., 2026). As widespread and rapid climate change alters habitats’ typical conditions (Loarie et al., 2009; Diffenbaugh and Field, 2013; IPCC, 2023) and increases the stochasticity and severity of extreme events (IPCC, 2023), species must respond to increasingly novel and unpredictable environments. However, responding to multifarious change (i.e., across multiple variables at once) is often more difficult than responding to each change in isolation (Polazzo et al., 2024). This is particularly true for the many species that are already at risk of extirpation or extinction and face increasing anthropogenic pressure (Tucker et al., 2018; Weststrate et al., 2024), especially those with strong site fidelity (Merkle et al., 2022). For example, Aikens et al. (2022) found that industrial activity reduced the ability of mule deer (*Odocoileus hemionus*) to migrate following forage availability (also see Xu et al., 2021), while Sawyer et al. (2019) showed that mule deer’s strong site fidelity prevents them from adapting to changing environments. In contrast, Ortega et al. (2023) demonstrated that mule deer adapt their migration to changes in phenology, suggesting that adaptability is likely population-dependent. For other examples of species’ responses to changes in environmental conditions and stochasticity, see Harris et al. (2018).

Quantifying trends in variance can also be helpful in broader disciplines, such as macroecology and landscape ecology. Changes in habitat variance over time can be indicators of ecological transitions and regime shifts (Carpenter and Brock, 2006; Bunting et al., 2016; Bjorndahl et al., 2022), since increases in stochasticity often precede a transition to an alternate stable state, whether the system be a habitat (Hastings et al., 2021) or a species at an evolutionary tipping point (Botero et al., 2015). The time series of mean NDVI and DENVar that we presented in Fig. 3 indicate two global-scale cycles of increased mean NDVI and decreased DENVar in the 1990s, but such cycles have not repeated since. Instead, changes post 2000 were relatively spatially consistent and latitudinally stratified, particularly following the El Ni3o event of 2015-2016. The magnitude and homogeneity of the changes hint at the possibility of large-scale shifts to alternate states. Still, four decades is not much data for estimating decadal trends, so further analysis is required to understand these complex dynamics. And while our work is limited to terrestrial systems, similar approaches may be applicable to freshwater systems (Chegoonian et al., 2023), or even coastal marine environments (Zheng and

DiGiacomo, 2017), and the open ocean (Hong et al., 2023), although the latter two would not use NDVI.

DENVar can also be relevant to a variety of industries as a proxy of variance in primary productivity (Pettorelli et al., 2005). In agriculture, remotely-sensed mean NDVI has been used to classify crop types (Zhang et al., 2022; Bhatti et al., 2024) and assess crop yield (Roznik et al., 2022). Accounting for spatiotemporal variance in NDVI may improve the accuracy of models of crop yield and harvest timing by accounting for the spatiotemporal variation in growth conditions. In land conservation and management (including forestry), mean NDVI can help assess a habitat’s regeneration stage following landscape disturbance (Pickell et al., 2016), including after fire (Lanorte et al., 2014; Valtonen et al., 2021; Zahabnazouri et al., 2025), logging (Chirici et al., 2020), and during other restoration efforts (Valtonen et al., 2021). Currently, little work has been done on the change in NDVI variance during or after recovery efforts, but quantifying such trends may help elucidate the recovery process. Lanorte et al. (2014) found that forests had greater Shannon entropy (Shannon, 1948) in NDVI prior to a fire than after, suggesting that temporal trends in NDVI variance may be useful metrics for assessing forests’ regeneration stage. When mean NDVI is not a good proxy of forest regeneration (e.g., Buma, 2012), the normalized burn ratio may be more appropriate (Chen et al., 2011). Still, variance in NDVI may help assess the landscape’s resilience to further disturbance and also distinguish between different types of vegetation, since canopy height and vegetation density impact the timing of snowmelt (Sun et al., 2022).

Irrespective of the field of application, the rasters presented here are most appropriate for processes that operate at similar spatiotemporal scale as the rasters (i.e., daily with a spatial resolution of ~ 0.1) or coarser. The problem of scale has been discussed at length in ecology (Levin, 1992; Frankham and Brook, 2004), and the importance of modeling processes at relevant scales has gained increasingly greater attention (Bro-Jørgensen et al., 2008; Nathan et al., 2022; Goodchild and Quattrochi, 2023; Pease, 2024). Importantly, analysts must carefully decide the scale and magnitude of what they (or the species of interest) consider an appreciable change, so that they can determine what counts as a (predictable) change in the mean and what is random variation around the mean. For example, the rasters are appropriate for studying the movement behaviour of ungulates, which can easily move several kilometers in a day, but they will likely have only weak (if any) correlation with the behaviour of individual insects, since they generally have much more limited vagility and lifespans. Still, the rasters may help understand the dynamics of insect populations, which operate at a larger and more stable scale than a single individual. Riotte-Lambert and Matthiopoulos (2020), Mezzini et al. (2025), and Oestreich et al. (2026) provide some useful background on the effects of spatiotemporal scale and unpredictability on animal movement. Furthermore, our estimates of mean NDVI and variance can be paired with finer-scale data to account for trends at various scales. For example, our rasters may inform about the seasonal phenology of a landscape, while data on fine-scale terrain characteristics, and hourly rainfall and temperature can help inform the smaller-scale trends.

In this paper, we have introduced a new estimate of environmental stochasticity, which we have named the Dynamic Estimate of NDVI Variance. We have illustrated some of its possible applications in ecological theory, conservation, forest regeneration, and agriculture. We showed that NDVI stochasticity has a concave-down relationship with mean NDVI across many biomes, and

that it also has a complex nonlinear relationship with several important ecological variables, including mean temperature, seasonal temperature range, total precipitation, and the diversity of endangered species. Still, quantifying spatiotemporal trends in environmental stochasticity remains a complex task, since it requires often prohibitive amounts of data, computation power, and time. Consequently, we hope analysts find our estimates of mean and variance in NDVI useful. As climate change and environmental stochasticity intensify rapidly over the coming decades and expose species to increasingly novel and unpredictable environments, we hope reasearchers and stakeholders begin to account for environmental stochasticity in their work, decision-making processes, and acknowledge that environmental stochasticity is a crucial property of many ecosystems.

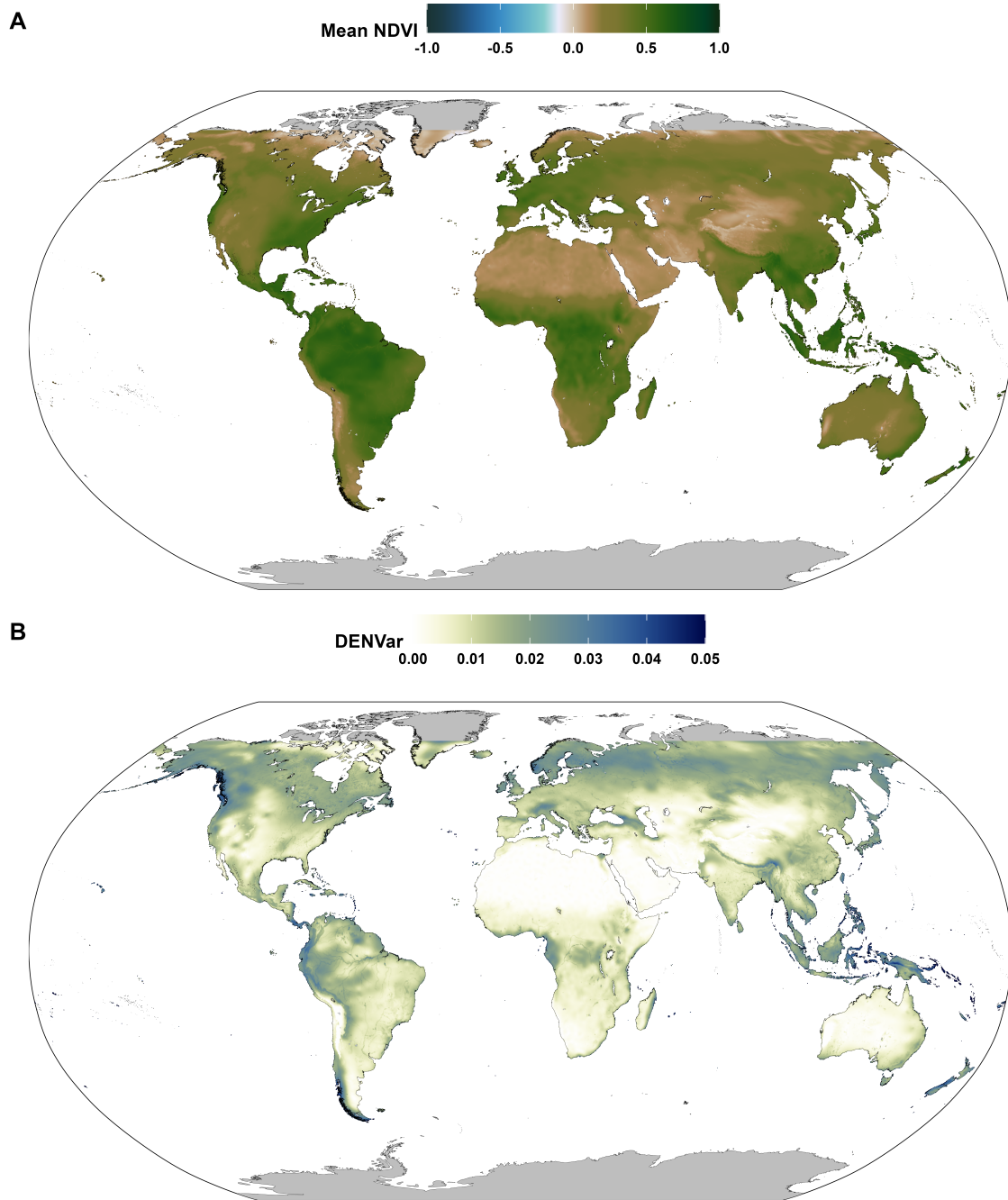


Figure 2: Estimated long-term mean NDVI (A) and DENVar (B).

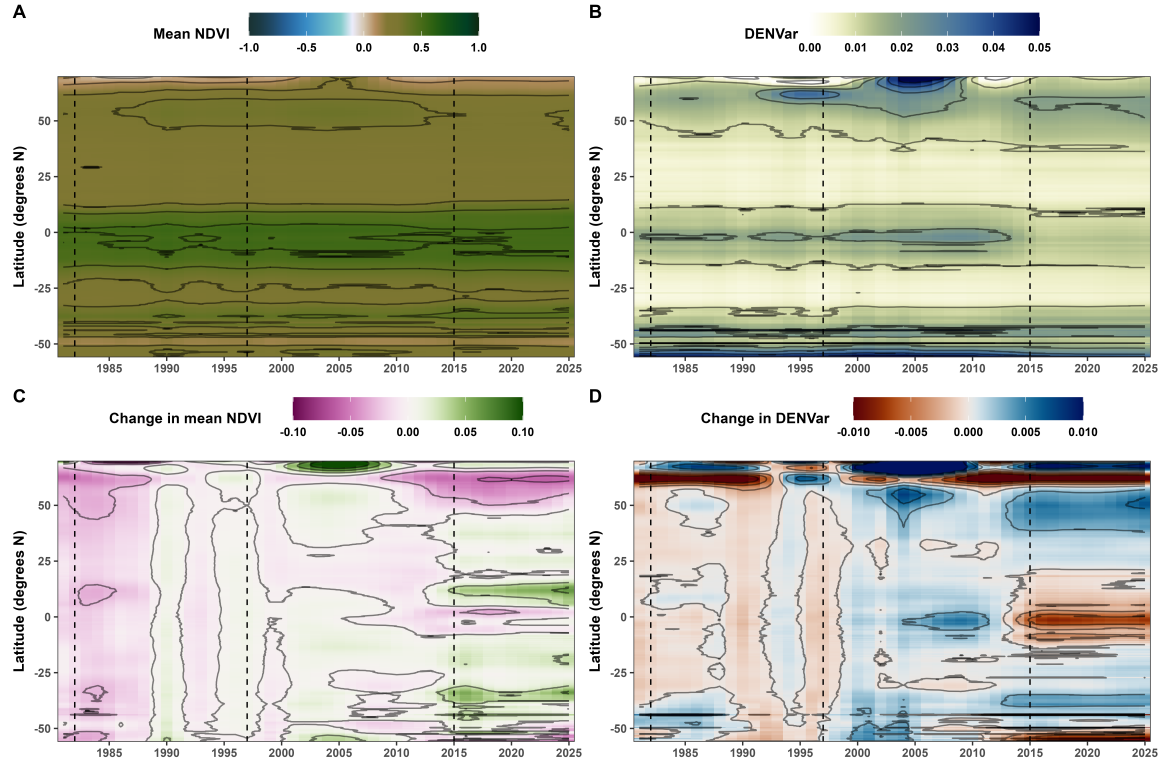


Figure 3: Surface plots of the change in mean NDVI and DENVar over the years. Panels A and B show the estimated averages, while panels C and D show the change in the averages relative to the 1991-2000 average. For ease of readability, DENVar values in panel B have been capped to a maximum of 0.05, while differences in panels C and D have been capped at absolute values of 0.1 and 0.01 ($= 0.1^2$), respectively. Vertical dashed lines indicate years with major El Niño events (Jia et al., 2025).

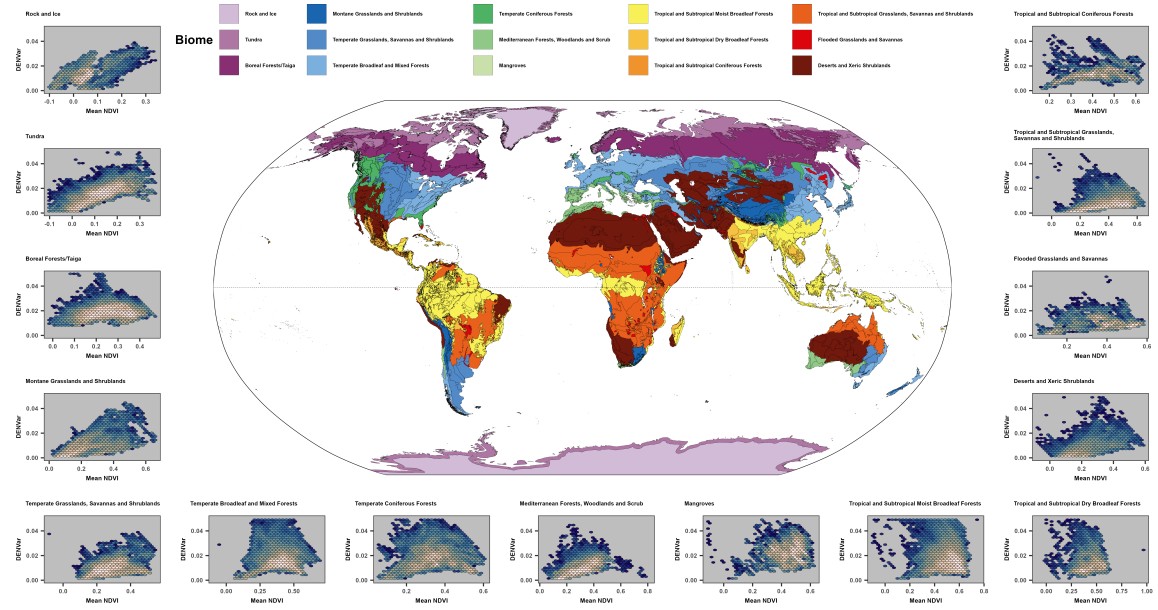


Figure 4: Map of terrestrial biomes along with hex plots of mean NDVI and DENVar for each biome. The fill scale of the hexagonal cells represents the log-10 number of points within the cell, scaled to the biome's total area, with lighter colors indicating higher density.

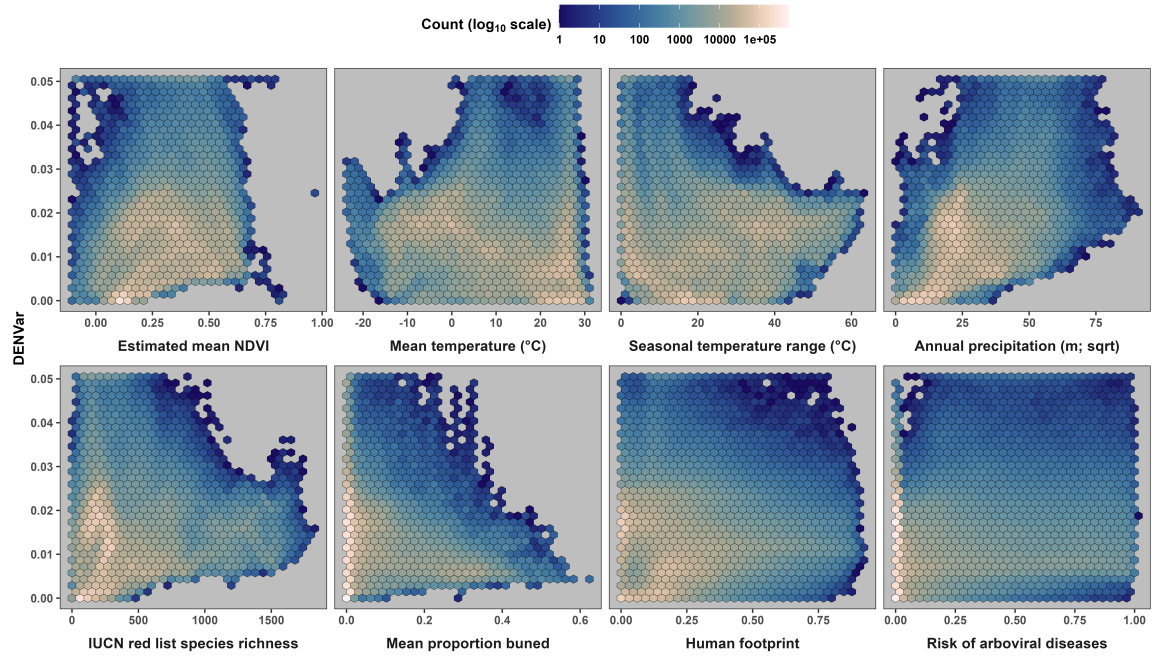


Figure 5: Hex plots of the relationship between DENVar and common ecological metrics. The fill of the hexagonal cells represents the log-10 number of points within the cell. Rowwise, the variables are: long-term mean NDVI (as estimated in this manuscript); mean annual temperature, seasonal temperature range, and annual precipitation (1970-2000) from the work of Fick and Hijmans (2017); IUCN Red List species richness (IUCN, 2025); mean proportion burned (1901-2020; Guo et al., 2025); the human footprint index of Keys et al. (2021); and the risk of dengue, chikungunya, Zika and yellow fever (Lim et al., 2025).

5 Other useful references

- **READ**: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data (<https://doi.org/10.1371/journal.pone.0041010>)
- Ecological and evolutionary consequences of changing seasonality: <https://www.science.org/doi/10.1126/science.ads>
- increase in variance before eutrophication in lakes: <https://doi.org/10.1002/lno.10355>
- uses ndvi variance: <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348>
- bro-jorgensen_using_2008: in the dry season preceding the rut topi antelope (*Damaliscus lunatus*) respond to forage availability at the large scale but not the fine scale
- keith_predicting_2008
- mueller_search_2008
- tian_evaluating_2015: NDVI changes across sensors
- wang_stochastic_2019
- pease_ecological_2024

Acknowledgments

The authors would like to thank Dr. Simon Wood for writing the custom Beta location-scale family even though it was not used in the final models and Dr. Gavin Simpson for providing useful feedback on early versions of the GAMs.

Author Contributions

Funding

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this article.

Code and Data Availability

All code is available on GitHub at <https://github.com/QuantitativeEcologyLab/ndvi-stochasticity>, including code for downloading the original NDVI rasters directly from the servers they are housed in. Most data are not included due to GitHub's size limitations, but all objects can be created using the code provided.

381 **Supplementary Materials**

- 382 • Appendix A: Color palettes used and input data.
- 383 • Appendix B: Sensitivity analyses.

References

- Briana Abrahms, Elliott L. Hazen, Ellen O. Aikens, Matthew S. Savoca, Jeremy A. Goldbogen, Steven J. Bograd, Michael G. Jacox, Ladd M. Irvine, Daniel M. Palacios, and Bruce R. Mate. Memory and resource tracking drive blue whale migrations. *Proceedings of the National Academy of Sciences*, 116(12):5582–5587, March 2019. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1819031116. URL <https://pnas.org/doi/full/10.1073/pnas.1819031116>.
- Ellen O. Aikens, Matthew J. Kauffman, Jerod A. Merkle, Samantha P. H. Dwinnell, Gary L. Fralick, and Kevin L. Monteith. The greenscape shapes surfing of resource waves in a large migratory herbivore. *Ecology Letters*, 20(6):741–750, June 2017. ISSN 1461-023X, 1461-0248. doi: 10.1111/ele.12772. URL <https://onlinelibrary.wiley.com/doi/10.1111/ele.12772>.
- Ellen O. Aikens, Kevin L. Monteith, Jerod A. Merkle, Samantha P. H. Dwinnell, Gary L. Fralick, and Matthew J. Kauffman. Drought reshuffles plant phenology and reduces the foraging benefit of green-wave surfing for a migratory ungulate. *Global Change Biology*, 26(8):4215–4225, August 2020. ISSN 1354-1013, 1365-2486. doi: 10.1111/gcb.15169. URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.15169>.
- Ellen O. Aikens, Teal B. Wyckoff, Hall Sawyer, and Matthew J. Kauffman. Industrial energy development decouples ungulate migration from the green wave. *Nature Ecology & Evolution*, 6(11):1733–1741, October 2022. ISSN 2397-334X. doi: 10.1038/s41559-022-01887-9. URL <https://www.nature.com/articles/s41559-022-01887-9>.
- Ellen O. Aikens, Jerod A. Merkle, Wenjing Xu, and Hall Sawyer. Pronghorn movements and mortality during extreme weather highlight the critical importance of connectivity. *Current Biology*, 35(8):1927–1934.e2, April 2025. ISSN 09609822. doi: 10.1016/j.cub.2025.03.010. URL <https://linkinghub.elsevier.com/retrieve/pii/S0960982225002957>.
- Sean C. Anderson, Trevor A. Branch, Andrew B. Cooper, and Nicholas K. Dulvy. Black-swan events in animal populations. *Proceedings of the National Academy of Sciences*, 114(12):3252–3257, March 2017. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1611525114. URL <http://www.pnas.org/lookup/doi/10.1073/pnas.1611525114>.
- Ramesh Arumugam, Frithjof Lutscher, and Frédéric Guichard. Tracking unstable states: ecosystem dynamics in a changing world. *Oikos*, 130(4):525–540, April 2021. ISSN 0030-1299, 1600-0706. doi: 10.1111/oik.08051. URL <https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/oik.08051>.
- Hylke E. Beck, Tim R. McVicar, Albert I.J.M. Van Dijk, Jaap Schellekens, Richard A.M. De Jeu, and L. Adrian Bruijnzeel. Global evaluation of four AVHRR–NDVI data sets: Intercomparison and assessment against Landsat imagery. *Remote Sensing of Environment*, 115(10):2547–2563, October 2011. ISSN 0034-4257. doi: 10.1016/j.rse.2011.05.012. URL <https://linkinghub.elsevier.com/retrieve/pii/S0034425711001957>.

- Be Beisner, Dt Haydon, and K. Cuddington. Alternative stable states in ecology. *Frontiers in Ecology and the Environment*, 1(7):376–382, September 2003. ISSN 1540-9295. doi: 10.1890/1540-9295(2003)001[0376:ASSIE]2.0.CO;2. URL [http://doi.wiley.com/10.1890/1540-9295\(2003\)001\[0376:ASSIE\]2.0.CO;2](http://doi.wiley.com/10.1890/1540-9295(2003)001[0376:ASSIE]2.0.CO;2).
- Muhammad Tousif Bhatti, Hammad Gilani, Muhammad Ashraf, Muhammad Shahid Iqbal, and Sarfraz Munir. Field validation of NDVI to identify crop phenological signatures. *Precision Agriculture*, 25(5):2245–2270, October 2024. ISSN 1385-2256, 1573-1618. doi: 10.1007/s11119-024-10165-6. URL <https://link.springer.com/10.1007/s11119-024-10165-6>.
- Judith A. Bjorndahl, Cale A.C. Gushulak, Stefano Mezzini, Gavin L. Simpson, Heather A. Haig, Peter R. Leavitt, and Kerri Finlay. Abrupt changes in the physical and biological structure of endorheic upland lakes due to 8-m lake-level variation during the 20th century. *Limnology and Oceanography*, 67(5):1022–1039, May 2022. ISSN 0024-3590, 1939-5590. doi: 10.1002/lno.12054. URL <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.12054>.
- Tomasz Borowik, Nathalie Pettorelli, Leif Sönnichsen, and Bogumiła Jędrzejewska. Normalized difference vegetation index (NDVI) as a predictor of forage availability for ungulates in forest and field habitats. *European Journal of Wildlife Research*, 59(5):675–682, October 2013. ISSN 1612-4642, 1439-0574. doi: 10.1007/s10344-013-0720-0. URL <http://link.springer.com/10.1007/s10344-013-0720-0>.
- Carlos A. Botero, Franz J. Weissing, Jonathan Wright, and Dustin R. Rubenstein. Evolutionary tipping points in the capacity to adapt to environmental change. *Proceedings of the National Academy of Sciences*, 112(1):184–189, January 2015. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1408589111. URL <https://pnas.org/doi/full/10.1073/pnas.1408589111>.
- Chloe Bracis and Thomas Mueller. Memory, not just perception, plays an important role in terrestrial mammalian migration. *Proceedings of the Royal Society B: Biological Sciences*, 284(1855):20170449, May 2017. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2017.0449. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2017.0449>.
- Jakob Bro-Jørgensen, Molly E. Brown, and Nathalie Pettorelli. Using the satellite-derived normalized difference vegetation index (NDVI) to explain ranging patterns in a lek-breeding antelope: the importance of scale. *Oecologia*, 158(1):177–182, November 2008. ISSN 0029-8549, 1432-1939. doi: 10.1007/s00442-008-1121-z. URL <http://link.springer.com/10.1007/s00442-008-1121-z>.
- James C Bull, Nicola J Pickup, Brian Pickett, Michael P Hassell, and Michael B Bonsall. Metapopulation extinction risk is increased by environmental stochasticity and assemblage complexity. *Proceedings of the Royal Society B: Biological Sciences*, 274(1606):87–96, January 2007. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2006.3691. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2006.3691>.
- Brian Buma. Evaluating the utility and seasonality of NDVI values for assessing post-disturbance recovery in a subalpine forest. *Environmental Monitoring and Assessment*, 184(6):3849–3860,

457 June 2012. ISSN 0167-6369, 1573-2959. doi: 10.1007/s10661-011-2228-y. URL [http://link.](http://link.springer.com/10.1007/s10661-011-2228-y)
 458 [springer.com/10.1007/s10661-011-2228-y](http://link.springer.com/10.1007/s10661-011-2228-y).

459 L. Bunting, P.R. Leavitt, G.L. Simpson, B. Wissel, K.R. Laird, B.F. Cumming, A. St. Amand,
 460 and D.R. Engstrom. Increased variability and sudden ecosystem state change in Lake Winnipeg,
 461 Canada, caused by 20th century agriculture: Lake Winnipeg Variability and State Change. *Limnol-*
 462 *ogy and Oceanography*, 61(6):2090–2107, November 2016. ISSN 00243590. doi: 10.1002/lno.10355.
 463 URL <https://onlinelibrary.wiley.com/doi/10.1002/lno.10355>.

464 Kendall L. Calhoun, Thomas Connor, Kaitlyn M. Gaynor, Amy Van Scoyoc, Alex McInturff,
 465 Samantha E.S. Kreling, and Justin S. Brashares. Movement behavior in a dominant ungulate
 466 underlies successful adjustment to a rapidly changing landscape following megafire. *Movement*
 467 *Ecology*, 12(1):53, July 2024. ISSN 2051-3933. doi: 10.1186/s40462-024-00488-4. URL [https:](https://movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-024-00488-4)
 468 [/movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-024-00488-4](https://movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-024-00488-4).

469 S. R. Carpenter and W. A. Brock. Rising variance: a leading indicator of ecological tran-
 470 sition. *Ecology Letters*, 9(3):311–318, March 2006. ISSN 1461-023X, 1461-0248. doi:
 471 10.1111/j.1461-0248.2005.00877.x. URL [https://onlinelibrary.wiley.com/doi/10.1111/j.](https://onlinelibrary.wiley.com/doi/10.1111/j.1461-0248.2005.00877.x)
 472 [1461-0248.2005.00877.x](https://onlinelibrary.wiley.com/doi/10.1111/j.1461-0248.2005.00877.x).

473 Amir M. Chegoonian, Nima Pahlevan, Kiana Zolfaghari, Peter R. Leavitt, John-Mark Davies, He-
 474 len M. Baulch, and Claude R. Duguay. Comparative Analysis of Empirical and Machine Learning
 475 Models for Chl *a* Extraction Using Sentinel-2 and Landsat OLI Data: Opportunities, Limitations,
 476 and Challenges. *Canadian Journal of Remote Sensing*, 49(1):2215333, July 2023. ISSN 0703-
 477 8992, 1712-7971. doi: 10.1080/07038992.2023.2215333. URL [https://www.tandfonline.com/](https://www.tandfonline.com/doi/full/10.1080/07038992.2023.2215333)
 478 [doi/full/10.1080/07038992.2023.2215333](https://www.tandfonline.com/doi/full/10.1080/07038992.2023.2215333).

479 Xuexia Chen, James E. Vogelmann, Matthew Rollins, Donald Ohlen, Carl H. Key, Limin Yang,
 480 Chengquan Huang, and Hua Shi. Detecting post-fire burn severity and vegetation recovery using
 481 multitemporal remote sensing spectral indices and field-collected composite burn index data in
 482 a ponderosa pine forest. *International Journal of Remote Sensing*, 32(23):7905–7927, December
 483 2011. ISSN 0143-1161, 1366-5901. doi: 10.1080/01431161.2010.524678. URL [https://www.](https://www.tandfonline.com/doi/full/10.1080/01431161.2010.524678)
 484 [tandfonline.com/doi/full/10.1080/01431161.2010.524678](https://www.tandfonline.com/doi/full/10.1080/01431161.2010.524678).

485 Luis-Miguel Chevin, Russell Lande, and Georgina M. Mace. Adaptation, Plasticity, and Extinction
 486 in a Changing Environment: Towards a Predictive Theory. *PLoS Biology*, 8(4):e1000357, April
 487 2010. ISSN 1545-7885. doi: 10.1371/journal.pbio.1000357. URL [https://dx.plos.org/10.1371/](https://dx.plos.org/10.1371/journal.pbio.1000357)
 488 [journal.pbio.1000357](https://dx.plos.org/10.1371/journal.pbio.1000357).

489 Gherardo Chirici, Francesca Giannetti, Erica Mazza, Saverio Francini, Davide Travaglini, Raf-
 490 faello Pegna, and Joanne C. White. Monitoring clearcutting and subsequent rapid recovery
 491 in Mediterranean coppice forests with Landsat time series. *Annals of Forest Science*, 77(2):
 492 40, June 2020. ISSN 1286-4560, 1297-966X. doi: 10.1007/s13595-020-00936-2. URL [https:](https://annforsci.biomedcentral.com/articles/10.1007/s13595-020-00936-2)
 493 [/annforsci.biomedcentral.com/articles/10.1007/s13595-020-00936-2](https://annforsci.biomedcentral.com/articles/10.1007/s13595-020-00936-2).

- 494 Fabio Crameri. Geodynamic diagnostics, scientific visualisation and StagLab 3.0. *Geoscientific Model*
 495 *Development*, 11(6):2541–2562, June 2018a. ISSN 1991-9603. doi: 10.5194/gmd-11-2541-2018.
 496 URL <https://gmd.copernicus.org/articles/11/2541/2018/>.
- 497 Fabio Crameri. Scientific colour-maps, June 2018b. URL <https://zenodo.org/record/1287763>.
 498 Language: en.
- 499 Fabio Crameri, Grace E. Shephard, and Philip J. Heron. The misuse of colour in science
 500 communication. *Nature Communications*, 11(1):5444, October 2020. ISSN 2041-1723. doi:
 501 10.1038/s41467-020-19160-7. URL <https://www.nature.com/articles/s41467-020-19160-7>.
- 502 Noah S. Diffenbaugh and Christopher B. Field. Changes in Ecologically Critical Terrestrial Climate
 503 Conditions. *Science*, 341(6145):486–492, August 2013. ISSN 0036-8075, 1095-9203. doi: 10.1126/
 504 science.1237123. URL <https://www.science.org/doi/10.1126/science.1237123>.
- 505 Andrea Falcón-Cortés, Denis Boyer, Evelyn Merrill, Jacqueline L. Frair, and Juan Manuel Morales.
 506 Hierarchical, Memory-Based Movement Models for Translocated Elk (*Cervus canadensis*). *Front-*
 507 *iers in Ecology and Evolution*, 9:702925, August 2021. ISSN 2296-701X. doi: 10.3389/fevo.2021.
 508 702925. URL <https://www.frontiersin.org/articles/10.3389/fevo.2021.702925/full>.
- 509 Xingwang Fan and Yuanbo Liu. A global study of NDVI difference among moderate-resolution
 510 satellite sensors. *ISPRS Journal of Photogrammetry and Remote Sensing*, 121:177–191, Novem-
 511 ber 2016. ISSN 09242716. doi: 10.1016/j.isprsjprs.2016.09.008. URL <https://linkinghub.elsevier.com/retrieve/pii/S0924271616303975>.
- 513 Christopher J. Fettig, Justin B. Runyon, Crystal S. Homicz, Patrick M. A. James, and Michael D.
 514 Ulyshen. Fire and Insect Interactions in North American Forests. *Current Forestry Reports*,
 515 8(4):301–316, August 2022. ISSN 2198-6436. doi: 10.1007/s40725-022-00170-1. URL <https://link.springer.com/10.1007/s40725-022-00170-1>.
- 517 Stephen E. Fick and Robert J. Hijmans. WorldClim 2: new 1-km spatial resolution climate surfaces
 518 for global land areas. *International Journal of Climatology*, 37(12):4302–4315, October 2017. ISSN
 519 0899-8418, 1097-0088. doi: 10.1002/joc.5086. URL <https://rmets.onlinelibrary.wiley.com/doi/10.1002/joc.5086>.
- 521 Belen Franch, Eric Vermote, Jean-Claude Roger, Emilie Murphy, Inbal Becker-Reshef, Chris Justice,
 522 Martin Claverie, Jyoteshwar Nagol, Ivan Csizar, Dave Meyer, Frederic Baret, Edward Masuoka,
 523 Robert Wolfe, and Sadashiva Devadiga. A 30+ Year AVHRR Land Surface Reflectance Climate
 524 Data Record and Its Application to Wheat Yield Monitoring. *Remote Sensing*, 9(3):296, March
 525 2017. ISSN 2072-4292. doi: 10.3390/rs9030296. URL <https://www.mdpi.com/2072-4292/9/3/296>.
- 527 Richard Frankham and Barry W. Brook. The importance of time scale in conservation biology and
 528 ecology. *Annales Zoologici Fennici*, 41:459–463, 2004. ISSN 0003-455X.

529 Michael J.M. Franklin, Richard E. Major, Michael Bedward, and Ross A. Bradstock. Relative
530 avian mobility linked to use of fire-affected resources in forested landscapes. *Forest Ecology and*
531 *Management*, 497:119484, October 2021. ISSN 03781127. doi: 10.1016/j.foreco.2021.119484. URL
532 <https://linkinghub.elsevier.com/retrieve/pii/S0378112721005739>.

533 Nicolas Frerebeau. *khroma: Colour Schemes for Scientific Data Visualization*. Université Bor-
534 deaux Montaigne, Pessac, France, 2024. doi: 10.5281/zenodo.1472077. URL [https://packages.](https://packages.tesselle.org/khroma/)
535 [tesselle.org/khroma/](https://packages.tesselle.org/khroma/).

536 Tak Fung, James P. O'Dwyer, and Ryan A. Chisholm. Quantifying species extinction risk under
537 temporal environmental variance. *Ecological Complexity*, 34:139–146, May 2018. ISSN 1476945X.
538 doi: 10.1016/j.ecocom.2017.09.006. URL [https://linkinghub.elsevier.com/retrieve/pii/](https://linkinghub.elsevier.com/retrieve/pii/S1476945X1730082X)
539 [S1476945X1730082X](https://linkinghub.elsevier.com/retrieve/pii/S1476945X1730082X).

540 Garmin International, Inc. and Esri. World Water Bodies, April 2025. URL [https://hub.arcgis.](https://hub.arcgis.com/content/e750071279bf450cbd510454a80f2e63/about)
541 [com/content/e750071279bf450cbd510454a80f2e63/about](https://hub.arcgis.com/content/e750071279bf450cbd510454a80f2e63/about).

542 Chris Geremia, Jerod A. Merkle, Daniel R. Eacker, Rick L. Wallen, P. J. White, Mark Hebblewhite,
543 and Matthew J. Kauffman. Migrating bison engineer the green wave. *Proceedings of the National*
544 *Academy of Sciences*, 116(51):25707–25713, December 2019. ISSN 0027-8424, 1091-6490. doi: 10.
545 1073/pnas.1913783116. URL <http://www.pnas.org/lookup/doi/10.1073/pnas.1913783116>.

546 Michael F. Goodchild and Dale A. Quattrochi. *Scale in Remote Sensing and GIS*. Routledge, New
547 York, 1 edition, January 2023. ISBN 978-0-203-74017-0. doi: 10.1201/9780203740170. URL
548 <https://www.taylorfrancis.com/books/9780203740170>.

549 Zhixuan Guo, Wei Li, Philippe Ciais, Stephen Sitch, Guido R. Van Der Werf, Simon P. K.
550 Bowering, Ana Bastos, Florent Mouillot, Jiaying He, Minxuan Sun, Lei Zhu, Xiaomeng Du,
551 Nan Wang, and Xiaomeng Huang. Reconstructed global monthly burned area maps from
552 1901 to 2020, February 2025. URL [https://essd.copernicus.org/articles/17/3599/2025/](https://essd.copernicus.org/articles/17/3599/2025/essd-17-3599-2025-discussion.html)
553 [essd-17-3599-2025-discussion.html](https://essd.copernicus.org/articles/17/3599/2025/essd-17-3599-2025-discussion.html).

554 R. M. B. Harris, L. J. Beaumont, T. R. Vance, C. R. Tozer, T. A. Remenyi, S. E. Perkins-
555 Kirkpatrick, P. J. Mitchell, A. B. Nicotra, S. McGregor, N. R. Andrew, M. Letnic, M. R.
556 Kearney, T. Wernberg, L. B. Hutley, L. E. Chambers, M.-S. Fletcher, M. R. Keatley, C. A.
557 Woodward, G. Williamson, N. C. Duke, and D. M. J. S. Bowman. Biological responses to
558 the press and pulse of climate trends and extreme events. *Nature Climate Change*, 8(7):
559 579–587, July 2018. ISSN 1758-678X, 1758-6798. doi: 10.1038/s41558-018-0187-9. URL
560 <https://www.nature.com/articles/s41558-018-0187-9>.

561 Alan Hastings, Karen C. Abbott, Kim Cuddington, Tessa B. Francis, Ying-Cheng Lai, Andrew
562 Morozov, Sergei Petrovskii, and Mary Lou Zeeman. Effects of stochasticity on the length and
563 behaviour of ecological transients. *Journal of The Royal Society Interface*, 18(180):20210257, July
564 2021. ISSN 1742-5662. doi: 10.1098/rsif.2021.0257. URL [https://royalsocietypublishing.](https://royalsocietypublishing.org/doi/10.1098/rsif.2021.0257)
565 [org/doi/10.1098/rsif.2021.0257](https://royalsocietypublishing.org/doi/10.1098/rsif.2021.0257).

566 G. Hmimina, E. Dufrêne, J.-Y. Pontailier, N. Delpierre, M. Aubinet, B. Caquet, A. De Grandcourt,
567 B. Burban, C. Flechard, A. Granier, P. Gross, B. Heinesch, B. Longdoz, C. Moureaux, J.-M.
568 Ourcival, S. Rambal, L. Saint André, and K. Soudani. Evaluation of the potential of MODIS
569 satellite data to predict vegetation phenology in different biomes: An investigation using ground-
570 based NDVI measurements. *Remote Sensing of Environment*, 132:145–158, May 2013. ISSN 0034-
571 4257. doi: 10.1016/j.rse.2013.01.010. URL [https://linkinghub.elsevier.com/retrieve/pii/
572 S0034425713000229](https://linkinghub.elsevier.com/retrieve/pii/S0034425713000229).

573 Jeffrey Hollister, Tarak Shah, Jakub Nowosad, Alec L. Robitaille, Marcus W. Beck, and Mike John-
574 son. *elevatr: Access Elevation Data from Various APIs*. 2023. doi: 10.5281/zenodo.8335450. URL
575 <https://github.com/jhollist/elevatr/>.

576 Zhongkun Hong, Di Long, Xingdong Li, Yiming Wang, Jianmin Zhang, Mohamed A. Hamouda, and
577 Mohamed M. Mohamed. A global daily gap-filled chlorophyll- *a* dataset in open oceans during
578 2001–2021 from multisource information using convolutional neural networks. *Earth System Sci-
579 ence Data*, 15(12):5281–5300, November 2023. ISSN 1866-3516. doi: 10.5194/essd-15-5281-2023.
580 URL <https://essd.copernicus.org/articles/15/5281/2023/>.

581 Sha Huang, Lina Tang, Joseph P. Hupy, Yang Wang, and Guofan Shao. A commentary review on
582 the use of normalized difference vegetation index (NDVI) in the era of popular remote sensing.
583 *Journal of Forestry Research*, 32(1):1–6, February 2021. ISSN 1007-662X, 1993-0607. doi: 10.
584 1007/s11676-020-01155-1. URL <https://link.springer.com/10.1007/s11676-020-01155-1>.

585 IPCC. *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to
586 the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge
587 University Press, 1 edition, July 2023. ISBN 978-1-009-15789-6. doi: 10.1017/9781009157896.
588 URL <https://www.cambridge.org/core/product/identifier/9781009157896/type/book>.

589 IUCN. The IUCN Red List of Threatened Species, 2025. URL <https://www.iucnredlist.org>.

590 Fan Jia, Wenju Cai, Tao Geng, Bolan Gan, Wenxiu Zhong, Lixin Wu, and Michael J. McPhaden.
591 Transition from multi-year La Niña to strong El Niño rare but increased under global warming.
592 *Science Bulletin*, 70(5):756–764, March 2025. ISSN 20959273. doi: 10.1016/j.scib.2024.12.034.
593 URL <https://linkinghub.elsevier.com/retrieve/pii/S2095927324009411>.

594 Heather E. Johnson, David D. Gustine, Trevor S. Golden, Layne G. Adams, Lincoln S. Parrett, Eliz-
595 abeth A. Lenart, and Perry S. Barboza. NDVI
596 exhibits mixed success in predicting spatiotemporal variation in caribou summer forage quality
597 and quantity. *Ecosphere*, 9(10):e02461, October 2018. ISSN 2150-8925, 2150-8925. doi: 10.1002/
598 ecs2.2461. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs2.2461>.

599 Torres João, Gonçalves João, Marcos Bruno, and Honrado João. Indicator-based assessment of
600 post-fire recovery dynamics using satellite NDVI time-series. *Ecological Indicators*, 89:199–212,
601 June 2018. ISSN 1470-160X. doi: 10.1016/j.ecolind.2018.02.008. URL [https://linkinghub.
602 elsevier.com/retrieve/pii/S1470160X18300852](https://linkinghub.elsevier.com/retrieve/pii/S1470160X18300852).

- Wayne S. Kendal. Taylor’s ecological power law as a consequence of scale invariant exponential dispersion models. *Ecological Complexity*, 1(3):193–209, September 2004. ISSN 1476945X. doi: 10.1016/j.ecocom.2004.05.001. URL <https://linkinghub.elsevier.com/retrieve/pii/S1476945X04000480>.
- Patrick W Keys, Elizabeth A Barnes, and Neil H Carter. A machine-learning approach to human footprint index estimation with applications to sustainable development. *Environmental Research Letters*, 16(4):044061, April 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/abe00a. URL <https://iopscience.iop.org/article/10.1088/1748-9326/abe00a>.
- Russell Lande. Risks of Population Extinction from Demographic and Environmental Stochasticity and Random Catastrophes. *The American Naturalist*, 142(6):911–927, 1993. ISSN 00030147, 15375323. URL <http://www.jstor.org/stable/2462690>.
- Antonio Lanorte, Rosa Lasaponara, Michele Lovallo, and Luciano Telesca. Fisher–Shannon information plane analysis of SPOT/VEGETATION Normalized Difference Vegetation Index (NDVI) time series to characterize vegetation recovery after fire disturbance. *International Journal of Applied Earth Observation and Geoinformation*, 26:441–446, February 2014. ISSN 15698432. doi: 10.1016/j.jag.2013.05.008. URL <https://linkinghub.elsevier.com/retrieve/pii/S0303243413000573>.
- Simon A. Levin. The Problem of Pattern and Scale in Ecology: The Robert H. MacArthur Award Lecture. *Ecology*, 73(6):1943–1967, December 1992. ISSN 0012-9658, 1939-9170. doi: 10.2307/1941447. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.2307/1941447>.
- Richard A. Levins. *Evolution in changing environments: some theoretical explorations*. Number 2 in Monographs in population biology. Princeton Univ. Press, Princeton, NJ, 3. printing edition, 1974. ISBN 978-0-691-08062-8.
- Ahyoung Lim, Freya M. Shearer, Kara Sewalk, David M. Pigott, Joseph Clarke, Azhar Ghouse, Ciara Judge, Hyolim Kang, Jane P. Messina, Moritz U. G. Kraemer, Katy A. M. Gaythorpe, William M. De Souza, Elaine O. Nsoesie, Michael Celone, Nuno Faria, Sadie J. Ryan, Ingrid B. Rabe, Diana P. Rojas, Simon I. Hay, John S. Brownstein, Nick Golding, and Oliver J. Brady. The overlapping global distribution of dengue, chikungunya, Zika and yellow fever. *Nature Communications*, 16(1):3418, April 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-58609-5. URL <https://www.nature.com/articles/s41467-025-58609-5>.
- Scott R. Loarie, Philip B. Duffy, Healy Hamilton, Gregory P. Asner, Christopher B. Field, and David D. Ackerly. The velocity of climate change. *Nature*, 462(7276):1052–1055, December 2009. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature08649. URL <https://www.nature.com/articles/nature08649>.
- Ramiro Logares and Martin Nuñez. Black Swans in ecology and evolution: The importance of improbable but highly influential events. *Ideas in Ecology and Evolution*, 2012. ISSN 19183178.

doi: 10.4033/iee.2012.5.5.n. URL <https://ojs.library.queensu.ca/index.php/IEE/article/view/4311>.

Paramesha Mallegowda, Ganesan Rengaiyan, Jayalakshmi Krishnan, and Madhura Niphadkar. Assessing Habitat Quality of Forest-Corridors through NDVI Analysis in Dry Tropical Forests of South India: Implications for Conservation. *Remote Sensing*, 7(2):1619–1639, February 2015. ISSN 2072-4292. doi: 10.3390/rs70201619. URL <https://www.mdpi.com/2072-4292/7/2/1619>.

Rekha Marcus and Michael J. Noonan. Changing variability is an overlooked aspect of protected area planning, October 2023. URL <http://biorxiv.org/lookup/doi/10.1101/2023.10.26.564062>.

Rekha Marcus, Stefano Mezzini, Dwija Desai, and Michael Noonan. Planning for a variable future: Environmental stochasticity, species richness, and protected areas in Canada, October 2025. URL <https://www.researchsquare.com/article/rs-7665738/v1>.

Bunkei Matsushita, Wei Yang, Jin Chen, Yuyichi Onda, and Guoyu Qiu. Sensitivity of the Enhanced Vegetation Index (EVI) and Normalized Difference Vegetation Index (NDVI) to Topographic Effects: A Case Study in High-density Cypress Forest. *Sensors*, 7(11):2636–2651, November 2007. ISSN 1424-8220. doi: 10.3390/s7112636. URL <https://www.mdpi.com/1424-8220/7/11/2636>.

Julian Merder, Robert A. Rigby, Andreas Mayr, Gillian Z. Heller, Thomas Kneib, Nikolaus Umlauf, Fernanda De Bastiani, Reto Stauffer, Jonathan D. Tonkin, Nikos Logothetis, Achim Zeileis, and Dimitrios M. Stasinopoulos. Distributional regression using generalized additive models for location, scale and shape. *Nature Reviews Methods Primers*, 6(1):49, July 2026. ISSN 2662-8449. doi: 10.1038/s43586-026-00498-z. URL <https://www.nature.com/articles/s43586-026-00498-z>.

Jerod A. Merkle, Kevin L. Monteith, Ellen O. Aikens, Matthew M. Hayes, Kent R. Hersey, Arthur D. Middleton, Brendan A. Oates, Hall Sawyer, Brandon M. Scurlock, and Matthew J. Kauffman. Large herbivores surf waves of green-up during spring. *Proceedings of the Royal Society B: Biological Sciences*, 283(1833):20160456, June 2016. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2016.0456. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2016.0456>.

Jerod A Merkle, Briana Abrahms, Jonathan B Armstrong, Hall Sawyer, Daniel P Costa, and Anna D Chalfoun. Site fidelity as a maladaptive behavior in the Anthropocene. *Frontiers in Ecology and the Environment*, 20(3):187–194, April 2022. ISSN 1540-9295, 1540-9309. doi: 10.1002/fee.2456. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/fee.2456>.

Stefano Mezzini, Christen H. Fleming, E. Patrícia Medici, and Michael J. Noonan. How resource abundance and resource stochasticity affect organisms’ range sizes. *Movement Ecology*, 13(1):20, March 2025. ISSN 2051-3933. doi: 10.1186/s40462-025-00546-5. URL <https://movementecologyjournal.biomedcentral.com/articles/10.1186/s40462-025-00546-5>.

Arthur D. Middleton, Jerod A. Merkle, Douglas E. McWhirter, John G. Cook, Rachel C. Cook, P. J. White, and Matthew J. Kauffman. Green-wave surfing increases fat gain in a migratory ungulate. *Oikos*, 127(7):1060–1068, July 2018. ISSN 00301299. doi: 10.1111/oik.05227. URL <https://onlinelibrary.wiley.com/doi/10.1111/oik.05227>.

- 676 Pedro R. Mutti, Paulo S. Lúcio, Vincent Dubreuil, and Bergson G. Bezerra. NDVI time series
677 stochastic models for the forecast of vegetation dynamics over desertification hotspots. *Inter-*
678 *national Journal of Remote Sensing*, 41(7):2759–2788, April 2020. ISSN 0143-1161, 1366-5901.
679 doi: 10.1080/01431161.2019.1697008. URL [https://www.tandfonline.com/doi/full/10.1080/](https://www.tandfonline.com/doi/full/10.1080/01431161.2019.1697008)
680 [01431161.2019.1697008](https://www.tandfonline.com/doi/full/10.1080/01431161.2019.1697008).
- 681 Ran Nathan, Christopher T. Monk, Robert Arlinghaus, Timo Adam, Josep Alós, Michael Assaf,
682 Henrik Baktoft, Christine E. Beardsworth, Michael G. Bertram, Allert I. Bijleveld, Tomas Brodin,
683 Jill L. Brooks, Andrea Campos-Candela, Steven J. Cooke, Karl Ø. Gjelland, Pratik R. Gupte,
684 Roi Harel, Gustav Hellström, Florian Jeltsch, Shaun S. Killen, Thomas Klefoth, Roland Langrock,
685 Robert J. Lennox, Emmanuel Lourie, Joah R. Madden, Yotam Orchan, Ine S. Pauwels, Milan Říha,
686 Manuel Roeleke, Ulrike E. Schlägel, David Shohami, Johannes Signer, Sivan Toledo, Ohad Vilks,
687 Samuel Westrelin, Mark A. Whiteside, and Ivan Jarić. Big-data approaches lead to an increased
688 understanding of the ecology of animal movement. *Science*, 375(6582):eabg1780, February 2022.
689 ISSN 0036-8075, 1095-9203. doi: 10.1126/science.abg1780. URL [https://www.science.org/](https://www.science.org/doi/10.1126/science.abg1780)
690 [doi/10.1126/science.abg1780](https://www.science.org/doi/10.1126/science.abg1780).
- 691 William K. Oestreich, Jenna E. Kohles, Briana Abrahms, Kelly J. Benoit-Bird, Andrew M. Berdahl,
692 John P. Ryan, and Dina K.N. Dechmann. Resource variability shapes the ecology of social infor-
693 mation and collective sensing. *Trends in Ecology & Evolution*, page S0169534726000261, March
694 2026. ISSN 01695347. doi: 10.1016/j.tree.2026.01.010. URL [https://linkinghub.elsevier.](https://linkinghub.elsevier.com/retrieve/pii/S0169534726000261)
695 [com/retrieve/pii/S0169534726000261](https://linkinghub.elsevier.com/retrieve/pii/S0169534726000261).
- 696 Masataka Okabe and Kei Ito. Color universal design (CUD): How to make figures and presentations
697 that are friendly to colorblind people, 2008. URL <https://jfly.uni-koeln.de/color/#pallet>.
- 698 David M. Olson, Eric Dinerstein, Eric D. Wikramanayake, Neil D. Burgess, George V. N. Powell,
699 Emma C. Underwood, Jennifer A. D’amico, Illanga Itoua, Holly E. Strand, John C. Morrison,
700 Colby J. Loucks, Thomas F. Allnutt, Taylor H. Ricketts, Yumiko Kura, John F. Lamoreux,
701 Wesley W. Wettengel, Prashant Hedao, and Kenneth R. Kassem. Terrestrial Ecoregions of the
702 World: A New Map of Life on Earth. *BioScience*, 51(11):933, 2001. ISSN 0006-3568. doi: 10.
703 1641/0006-3568(2001)051[0933:teotwa]2.0.co;2. URL [https://academic.oup.com/bioscience/](https://academic.oup.com/bioscience/article/51/11/933-938/227116)
704 [article/51/11/933-938/227116](https://academic.oup.com/bioscience/article/51/11/933-938/227116).
- 705 Anna C. Ortega, Ellen O. Aikens, Jerod A. Merkle, Kevin L. Monteith, and Matthew J. Kauffman.
706 Migrating mule deer compensate en route for phenological mismatches. *Nature Communications*,
707 14(1):2008, April 2023. ISSN 2041-1723. doi: 10.1038/s41467-023-37750-z. URL [https://www.](https://www.nature.com/articles/s41467-023-37750-z)
708 [nature.com/articles/s41467-023-37750-z](https://www.nature.com/articles/s41467-023-37750-z).
- 709 Brent S. Pease. Ecological scales of effect vary across space and time. *Ecography*, 2024(8):e07163,
710 August 2024. ISSN 0906-7590, 1600-0587. doi: 10.1111/ecog.07163. URL [https://nsojournals.](https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/ecog.07163)
711 [onlinelibrary.wiley.com/doi/10.1111/ecog.07163](https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/ecog.07163).
- 712 Eric J. Pedersen, David L. Miller, Gavin L. Simpson, and Noam Ross. Hierarchical generalized

additive models in ecology: an introduction with mgcv. *PeerJ*, 7:e6876, May 2019. ISSN 2167-8359. doi: 10.7717/peerj.6876. URL <https://peerj.com/articles/6876>.

N Pettorelli, S Ryan, T Mueller, N Bunnefeld, B Jedrzejewska, M Lima, and K Kausrud. The Normalized Difference Vegetation Index (NDVI): unforeseen successes in animal ecology. *Climate Research*, 46(1):15–27, January 2011. ISSN 0936-577X, 1616-1572. doi: 10.3354/cr00936. URL <http://www.int-res.com/abstracts/cr/v46/n1/p15-27/>.

Nathalie Pettorelli, Jon Olav Vik, Atle Mysterud, Jean-Michel Gaillard, Compton J. Tucker, and Nils Chr. Stenseth. Using the satellite-derived NDVI to assess ecological responses to environmental change. *Trends in Ecology & Evolution*, 20(9):503–510, September 2005. ISSN 01695347. doi: 10.1016/j.tree.2005.05.011. URL <https://linkinghub.elsevier.com/retrieve/pii/S016953470500162X>.

Paul D. Pickell, Txomin Hermosilla, Ryan J. Frazier, Nicholas C. Coops, and Michael A. Wulder. Forest recovery trends derived from Landsat time series for North American boreal forests. *International Journal of Remote Sensing*, 37(1):138–149, January 2016. ISSN 0143-1161, 1366-5901. doi: 10.1080/2150704X.2015.1126375. URL <https://www.tandfonline.com/doi/full/10.1080/2150704X.2015.1126375>.

Leo Polansky, Werner Kilian, and George Wittemyer. Elucidating the significance of spatial memory on movement decisions by African savannah elephants using state-space models. *Proceedings of the Royal Society B: Biological Sciences*, 282(1805):20143042, April 2015. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2014.3042. URL <https://royalsocietypublishing.org/doi/10.1098/rspb.2014.3042>.

Francesco Polazzo, Romana Limberger, Frank Pennekamp, Samuel R. P.-J. Ross, Gavin L. Simpson, and Owen L. Petchey. Measuring the Response Diversity of Ecological Communities Experiencing Multifarious Environmental Change. *Global Change Biology*, 30(11):e17594, November 2024. ISSN 1354-1013, 1365-2486. doi: 10.1111/gcb.17594. URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.17594>.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2025. URL <https://www.R-project.org/>.

Nathan Ranc, Francesca Cagnacci, and Paul R. Moorcroft. Memory drives the formation of animal home ranges: Evidence from a reintroduction. *Ecology Letters*, 25(4):716–728, April 2022. ISSN 1461-023X, 1461-0248. doi: 10.1111/ele.13869. URL <https://onlinelibrary.wiley.com/doi/10.1111/ele.13869>.

Colleen E. Reid, Laura D. Kubzansky, Jiayue Li, Jessie L. Shmool, and Jane E. Clougherty. It’s not easy assessing greenness: A comparison of NDVI datasets and neighborhood types and their associations with self-rated health in New York City. *Health & Place*, 54:92–101, November 2018. ISSN 1353-8292. doi: 10.1016/j.healthplace.2018.09.005. URL <https://linkinghub.elsevier.com/retrieve/pii/S1353829218301151>.

Gregory J. M. Rickbeil, Jerod A. Merkle, Greg Anderson, M. Paul Atwood, Jon P. Beckmann,
Eric K. Cole, Alyson B. Courtemanch, Sarah Dewey, David D. Gustine, Matthew J. Kauffman,
Douglas E. McWhirter, Tony Mong, Kelly Proffitt, Patrick J. White, and Arthur D. Middleton.
Plasticity in elk migration timing is a response to changing environmental conditions. *Global
Change Biology*, 25(7):2368–2381, July 2019. ISSN 1354-1013, 1365-2486. doi: 10.1111/gcb.14629.
URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.14629>.

Louise Riotte-Lambert and Jason Matthiopoulos. Environmental Predictability as a Cause and
Consequence of Animal Movement. *Trends in Ecology & Evolution*, 35(2):163–174, February
2020. ISSN 01695347. doi: 10.1016/j.tree.2019.09.009. URL [https://linkinghub.elsevier.
com/retrieve/pii/S0169534719302885](https://linkinghub.elsevier.com/retrieve/pii/S0169534719302885).

Matteo Rizzuto, Shawn J. Leroux, Eric Vander Wal, Isabella C. Richmond, Travis R. Heckford,
Juliana Balluffi-Fry, and Yolanda F. Wiersma. Forage stoichiometry predicts the home range
size of a small terrestrial herbivore. *Oecologia*, 197(2):327–338, October 2021. ISSN 0029-8549,
1432-1939. doi: 10.1007/s00442-021-04965-0. URL [https://link.springer.com/10.1007/
s00442-021-04965-0](https://link.springer.com/10.1007/s00442-021-04965-0).

Mitchell Roznik, Milton Boyd, and Lysa Porth. Improving crop yield estimation by applying higher
resolution satellite NDVI imagery and high-resolution cropland masks. *Remote Sensing Applica-
tions: Society and Environment*, 25:100693, January 2022. ISSN 23529385. doi: 10.1016/j.rsase.
2022.100693. URL <https://linkinghub.elsevier.com/retrieve/pii/S2352938522000015>.

Hall Sawyer, Jerod A. Merkle, Arthur D. Middleton, Samantha P. H. Dwinnell, and Kevin L.
Monteith. Migratory plasticity is not ubiquitous among large herbivores. *Journal of Animal
Ecology*, 88(3):450–460, March 2019. ISSN 0021-8790, 1365-2656. doi: 10.1111/1365-2656.12926.
URL <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.12926>.

C. E. Shannon. A Mathematical Theory of Communication. *Bell System Technical Journal*, 27
(3):379–423, July 1948. ISSN 00058580. doi: 10.1002/j.1538-7305.1948.tb01338.x. URL [https:
//ieeexplore.ieee.org/document/6773024](https://ieeexplore.ieee.org/document/6773024).

Gavin L. Simpson. Modelling Palaeoecological Time Series Using Generalised Additive Models.
Frontiers in Ecology and Evolution, 6:149, October 2018. ISSN 2296-701X. doi: 10.3389/fevo.
2018.00149. URL <https://www.frontiersin.org/article/10.3389/fevo.2018.00149/full>.

N.T. Son, C.F. Chen, C.R. Chen, V.Q. Minh, and N.H. Trung. A comparative analysis of multitem-
poral MODIS EVI and NDVI data for large-scale rice yield estimation. *Agricultural and Forest
Meteorology*, 197:52–64, October 2014. ISSN 0168-1923. doi: 10.1016/j.agrformet.2014.06.007.
URL <https://linkinghub.elsevier.com/retrieve/pii/S016819231400149X>.

Saurabh Steixner-Kumar and Jan Gläser. Strategies for navigating a dynamic world. *Science*, 369
(6507):1056–1057, August 2020. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.abd7258. URL
<https://www.science.org/doi/10.1126/science.abd7258>.

- 786 D. W. Stephens and Eric L. Charnov. Optimal foraging: Some simple stochastic models. *Behavioral*
787 *Ecology and Sociobiology*, 10(4):251–263, July 1982. ISSN 0340-5443, 1432-0762. doi: 10.1007/
788 BF00302814. URL <http://link.springer.com/10.1007/BF00302814>.
- 789 Ning Sun, Hongxiang Yan, Mark S. Wigmosta, Jessica Lundquist, Susan Dickerson-Lange, and
790 Tian Zhou. Forest Canopy Density Effects on Snowpack Across the Climate Gradients of the
791 Western United States Mountain Ranges. *Water Resources Research*, 58(1):e2020WR029194, Jan-
792 uary 2022. ISSN 0043-1397, 1944-7973. doi: 10.1029/2020WR029194. URL <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2020WR029194>.
- 794 Bernt-Erik Sæther. Environmental stochasticity and population dynamics of large herbivores: a
795 search for mechanisms. *Trends in Ecology & Evolution*, 12(4):143–149, April 1997. ISSN 0169-
796 5347. doi: 10.1016/S0169-5347(96)10068-9. URL [https://doi.org/10.1016/S0169-5347\(96\)](https://doi.org/10.1016/S0169-5347(96)10068-9)
797 10068-9.
- 798 L. R. Taylor. Aggregation, Variance and the Mean. *Nature*, 189(4766):732–735, March 1961.
799 ISSN 0028-0836, 1476-4687. doi: 10.1038/189732a0. URL [https://www.nature.com/articles/](https://www.nature.com/articles/189732a0)
800 189732a0.
- 801 Martha Torstenson and Allison K. Shaw. Strength of seasonality and type of migratory cue determine
802 the fitness consequences of changing phenology for migratory animals. *Oikos*, 2025(1):e10862,
803 January 2025. ISSN 0030-1299, 1600-0706. doi: 10.1111/oik.10862. URL <https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/oik.10862>.
- 805 Marlee A. Tucker, Katrin Böhning-Gaese, William F. Fagan, John M. Fryxell, Bram Van Moorter,
806 Susan C. Alberts, Abdullahi H. Ali, Andrew M. Allen, Nina Attias, Tal Avgar, Hattie Bartlam-
807 Brooks, Buuveibaatar Bayarbaatar, Jerrold L. Belant, Alessandra Bertassoni, Dean Beyer, Laura
808 Bidner, Floris M. van Beest, Stephen Blake, Niels Blaum, Chloe Bracis, Danielle Brown,
809 P. J. Nico de Bruyn, Francesca Cagnacci, Justin M. Calabrese, Constança Camilo-Alves, Si-
810 mon Chamailé-Jammes, Andre Chiaradia, Sarah C. Davidson, Todd Dennis, Stephen DeSte-
811 fano, Duane Diefenbach, Iain Douglas-Hamilton, Julian Fennessy, Claudia Fichtel, Wolfgang
812 Fiedler, Christina Fischer, Ilya Fischhoff, Christen H. Fleming, Adam T. Ford, Susanne A.
813 Fritz, Benedikt Gehr, Jacob R. Goheen, Eliezer Gurarie, Mark Hebblewhite, Marco Heurich,
814 A. J. Mark Hewison, Christian Hof, Edward Hurme, Lynne A. Isbell, René Janssen, Flor-
815 ian Jeltsch, Petra Kaczensky, Adam Kane, Peter M. Kappeler, Matthew Kauffman, Roland
816 Kays, Duncan Kimuyu, Flavia Koch, Bart Kranstauber, Scott LaPoint, Peter Leimgruber, John
817 D. C. Linnell, Pascual López-López, A. Catherine Markham, Jenny Mattisson, Emilia Patri-
818 cia Medici, Ugo Mellone, Evelyn Merrill, Guilherme de Miranda Mourão, Ronaldo G. Morato,
819 Nicolas Morellet, Thomas A. Morrison, Samuel L. Díaz-Muñoz, Atle Mysterud, Dejid Nandintset-
820 seg, Ran Nathan, Aidin Niamir, John Odden, Robert B. O’Hara, Luiz Gustavo R. Oliveira-
821 Santos, Kirk A. Olson, Bruce D. Patterson, Rogerio Cunha de Paula, Luca Pedrotti, Björn
822 Reineking, Martin Rimmler, Tracey L. Rogers, Christer Moe Rolandsen, Christopher S. Rosen-
823 berry, Daniel I. Rubenstein, Kamran Safi, Sonia Saïd, Nir Sapir, Hall Sawyer, Niels Martin

824 Schmidt, Nuria Selva, Agnieszka Sergiel, Enkhtuvshin Shiilegdamba, João Paulo Silva, Navin-
825 der Singh, Erling J. Solberg, Orr Spiegel, Olav Strand, Siva Sundaresan, Wiebke Ullmann,
826 Ulrich Voigt, Jake Wall, David Wattles, Martin Wikelski, Christopher C. Wilmers, John W.
827 Wilson, George Wittemyer, Filip Zięba, Tomasz Zwijacz-Kozica, and Thomas Mueller. Mov-
828 ing in the Anthropocene: Global reductions in terrestrial mammalian movements. *Science*, 359
829 (6374):466–469, January 2018. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.aam9712. URL
830 <https://www.sciencemag.org/lookup/doi/10.1126/science.aam9712>.

831 United States National Aeronautics and Space Administration. Level-1 and Atmosphere Archive &
832 Distribution System Distributed Active Archive Center, 2025. URL <https://ladsweb.modaps.eosdis.nasa.gov/archive/allData>.

834 United States National Centers for Environmental Information (NCEI) and United States National
835 Aeronautics and Space Administration. NCEI Database, 2025. URL <https://ncei.noaa.gov/dataver>.

837 Anu Valtonen, Eveliina Korkiatupa, Sille Holm, Geoffrey M. Malinga, and Ryosuke Nakadai. Re-
838 motely sensed vegetation greening along a restoration gradient of a tropical forest, Kibale National
839 Park, Uganda. *Land Degradation & Development*, 32(18):5166–5177, December 2021. ISSN 1085-
840 3278, 1099-145X. doi: 10.1002/ldr.4096. URL <https://onlinelibrary.wiley.com/doi/10.1002/ldr.4096>.

842 Eric Vermote and NOAA CDR Program. NOAA Climate Data Record (CDR) of AVHRR Normal-
843 ized Difference Vegetation Index (NDVI), Version 5, 2018. URL <https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01558/html>.

845 Eric Vermote and NOAA CDR Program. NOAA Climate Data Record (CDR) of VIIRS Normal-
846 ized Difference Vegetation Index (NDVI), Version 1, 2022. URL <https://www.ncei.noaa.gov/metadata/geoportal/rest/metadata/item/gov.noaa.ncdc:C01677/html>.

848 Dayna K. Weststrate, Aimee Chhen, Stefano Mezzini, Kirk Safford, and Michael J. Noonan. How
849 climate change and population growth will shape attendance and human-wildlife interactions at
850 British Columbia parks. *Journal of Sustainable Tourism*, pages 1–15, March 2024. ISSN 0966-
851 9582, 1747-7646. doi: 10.1080/09669582.2024.2331228. URL <https://www.tandfonline.com/doi/full/10.1080/09669582.2024.2331228>.

853 Simon N. Wood. *Generalized additive models: an introduction with R*. Chapman & Hall/CRC texts
854 in statistical science. CRC Press/Taylor & Francis Group, Boca Raton, second edition edition,
855 2017. ISBN 978-1-4987-2833-1.

856 Simon N. Wood and Matteo Fasiolo. A Generalized Feller-Schall Method for Smoothing Parameter
857 Optimization with Application to Tweedie Location, Scale and Shape Models. *Biometrics*, 73
858 (4):1071–1081, December 2017. ISSN 0006-341X, 1541-0420. doi: 10.1111/biom.12666. URL
859 <https://academic.oup.com/biometrics/article/73/4/1071-1081/7537785>.

- Simon N. Wood, Yannig Goude, and Simon Shaw. Generalized additive models for large data sets. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 64(1):139–155, January 2015. ISSN 00359254. doi: 10.1111/rssc.12068. URL <https://onlinelibrary.wiley.com/doi/10.1111/rssc.12068>.
- Simon N. Wood, Zheyuan Li, Gavin Shaddick, and Nicole H. Augustin. Generalized Additive Models for Gigadata: Modeling the U.K. Black Smoke Network Daily Data. *Journal of the American Statistical Association*, 112(519):1199–1210, July 2017. ISSN 0162-1459, 1537-274X. doi: 10.1080/01621459.2016.1195744. URL <https://www.tandfonline.com/doi/full/10.1080/01621459.2016.1195744>.
- Wenjing Xu, Kristin Barker, Avery Shawler, Amy Van Scoyoc, Justine A. Smith, Thomas Mueller, Hall Sawyer, Chelsea Andreozzi, Owen R. Bidder, Harshad Karandikar, Steffen Mumme, Elizabeth Templin, and Arthur D. Middleton. The plasticity of ungulate migration in a changing world. *Ecology*, 102(4):e03293, April 2021. ISSN 0012-9658, 1939-9170. doi: 10.1002/ecy.3293. URL <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.3293>.
- Somayeh Zahabnazouri, Patrick Belmont, Scott David, Peter E. Wigand, Mario Elia, and Domenico Capolongo. Detecting Burn Severity and Vegetation Recovery After Fire Using dNBR and dNDVI Indices: Insight from the Bosco Difesa Grande, Gravina in Southern Italy. *Sensors*, 25(10):3097, May 2025. ISSN 1424-8220. doi: 10.3390/s25103097. URL <https://www.mdpi.com/1424-8220/25/10/3097>.
- Lifu Zhang, Liaoran Gao, Changping Huang, Nan Wang, Sa Wang, Mingyuan Peng, Xia Zhang, and Qingxi Tong. Crop classification based on the spectrottemporal signature derived from vegetation indices and accumulated temperature. *International Journal of Digital Earth*, 15(1):626–652, December 2022. ISSN 1753-8947, 1753-8955. doi: 10.1080/17538947.2022.2036832. URL <https://www.tandfonline.com/doi/full/10.1080/17538947.2022.2036832>.
- Xueyan Zhang, Yunfeng Hu, Dafang Zhuang, Yongqing Qi, and Xin Ma. NDVI spatial pattern and its differentiation on the Mongolian Plateau. *Journal of Geographical Sciences*, 19(4):403–415, August 2009. ISSN 1009-637X, 1861-9568. doi: 10.1007/s11442-009-0403-7. URL <http://link.springer.com/10.1007/s11442-009-0403-7>.
- Guangming Zheng and Paul M. DiGiacomo. Remote sensing of chlorophyll-a in coastal waters based on the light absorption coefficient of phytoplankton. *Remote Sensing of Environment*, 201:331–341, November 2017. ISSN 00344257. doi: 10.1016/j.rse.2017.09.008. URL <https://linkinghub.elsevier.com/retrieve/pii/S0034425717304200>.